Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy turning small businesses' expansion questions into site analyses; found a $1.5M expansion opportunity.
- Estimated how much revenue a new location would add with causal ML (Double ML, causal forests) on 12 years of county data, validated with difference-in-differences and a synthetic-data recovery test.
- Built revenue projections and Monte Carlo risk models (probability of loss, VaR, CVaR) so owners saw the downside of a site before committing capital.
- Engineered Python ETL pipelines with automated data-quality checks and 150+ unit tests for Census, traffic, and competitor data, cutting market analysis turnaround by 91% (weeks to days).
- Designed a site-scoring model combining demographic, traffic, competitive-density, and consumer-behavior features, explaining results and uncertainty ranges to non-technical owners in written reports and maps.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, engineering, and behavioral science leadership, setting OKRs and translating business questions into models.
- Applied predictive, segmentation, and uplift models to millions of HIPAA-protected (PHI) Medicare/Medicaid records, cutting ad spend by $10K per campaign.
- Designed and standardized cross-team A/B testing and causal inference frameworks for direct-mail campaigns, applying CUPED to reduce metric variance by 20%.
- Owned monthly monitoring of drift, performance, and media mix forecast-vs-actual for up to 10 production models, scheduling Databricks jobs that refreshed the data and metrics feeding retraining and scoring.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory consumption by 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined metrics, evaluation, and drift monitoring for production search and regression models; mentored the team on ML.
- Built SQL training tables grouping bulk orders into item-level shipping costs, then deployed a regression-based shipping-price API in C# on Azure, scaling to 10,000+ daily requests.
- Reduced search customer dissatisfaction by 12% with product and engineering partners by building NLP relevance features: custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built XGBoost classifiers on millions of healthcare records with hundreds of attributes, lifting response rates 3%; explained drivers with SHAP.
- Automated reporting pipelines and Power BI dashboards for stakeholders with Python, SQL, and Snowflake, removing 60 minutes of manual work daily.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas, focusing campaign resources on high-opportunity regions.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in mathematical optimization, functional analysis, and operator theory for doctoral dissertation research.
- Taught mathematics and statistics to classes of 75-200 students and mentored 100+ students in summer programs, achieving a 95% pass rate.

TECHNICAL PROJECTS

CaLaun — *AI-Powered Calculus Tutor (LLM Application)* github.com/calaun-learning/calaun

- Built and deployed a generative AI tutor that grounds a self-hosted Llama 3.2 LLM in SymPy-verified solution steps, scoping each prompt to one step so the model explains but never computes; 400+ weekly users, 90+ automated tests.

Medicare Claims Intelligence github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse over 790K CMS Medicare claims and a provider fraud (FWA) detector that flagged planted billing schemes with 81% precision.
- Cut forecast error 41% vs. a seasonal benchmark (ARIMA, ETS, Prophet); built a SHAP-explained XGBoost model that finds 46% more high-cost members than baseline (MLflow) and a RAG assistant with PHI guardrails.

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show where LTV = ARPU / churn fails, estimating customer lifetime value from empirical survival curves.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, statsmodels, PyTorch, GeoPandas), SQL, C#, Bash, PowerShell

Machine Learning & AI: Regression, Classification, Clustering, XGBoost, Anomaly Detection, Time Series Forecasting (ARIMA, ETS, Prophet), Survival Analysis, Uplift Modeling, Media Mix Modeling, SHAP, NLP, LLMs (Llama), RAG, Prompt Engineering, Feature Engineering, Model Monitoring & Drift

Statistics & Experimentation: Hypothesis Testing, A/B Testing, CUPED, Power Analysis, Causal Inference, Double ML, Causal Forests, Difference-in-Differences, CATE, Monte Carlo Simulation, Optimization

Data Engineering & Cloud: Spark, Databricks (Workflows), Snowflake, dbt, MLflow, AWS, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT Pipelines, Data Quality Checks, Unit Testing, Git, AI-Assisted Development (Claude Code)

Visualization & Communication: Power BI, Tableau, Excel, Matplotlib, Plotly, Dashboards, Stakeholder Presentations, Mentoring